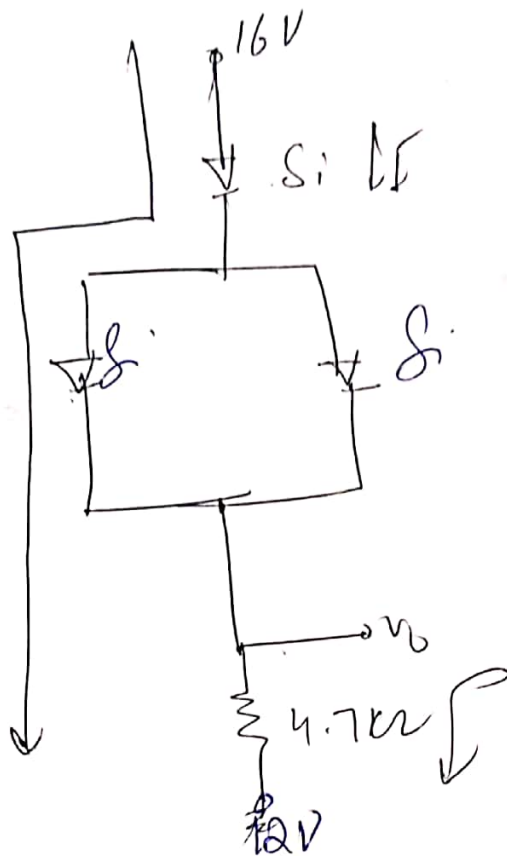


Q6.



$$16V > 12V$$

$$16 - 0.7 - 0.7 - 4.7 I = 12V$$

$$V_o = 12V$$

$$I = 0.55 \text{ mA}$$

$$V_o - 4.7 I = 12V$$

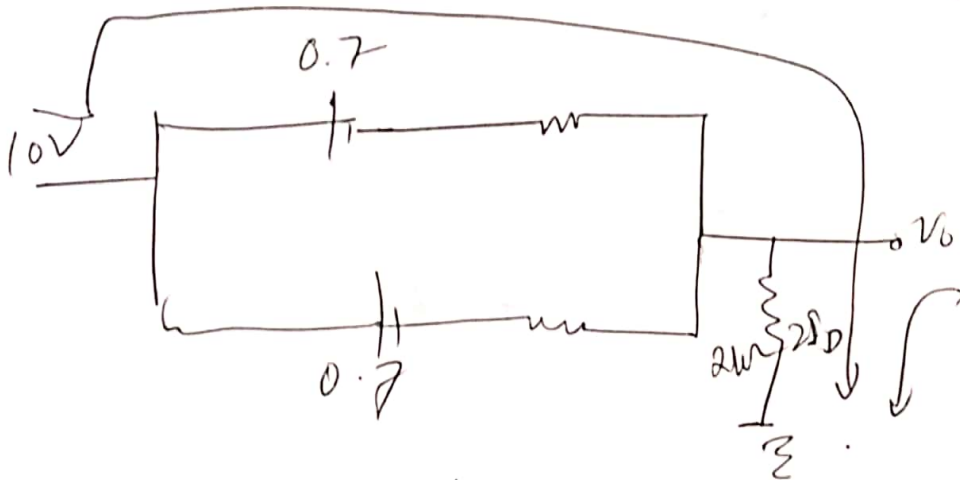
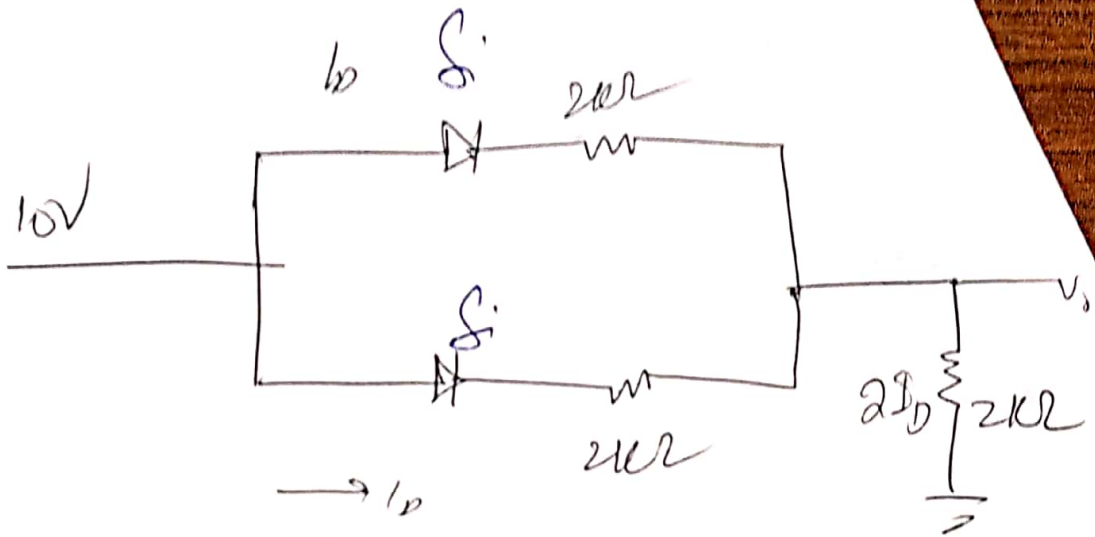
$$V_o = 12 + (4.7)(0.55 \text{ mA})$$

$$V_o = 14.6 \text{ volts}$$

3

Q5

1-



$$10 - 0.7 - 2I_D - 4I_D = 0$$

$$6I_D = 9.3$$

$$I_D = 1.5 \text{ mA}$$

Ans

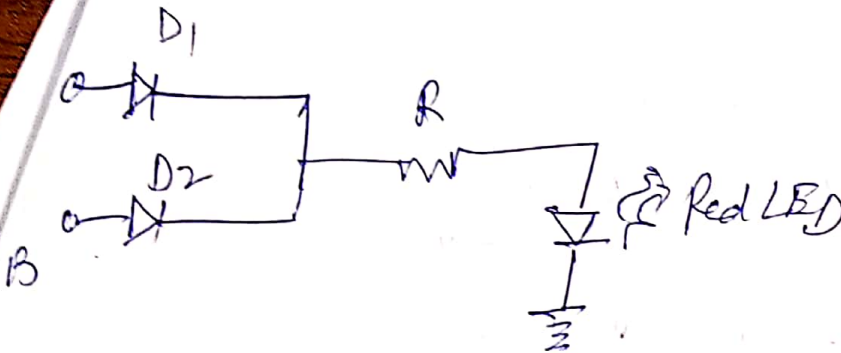
$$V_o - I_D R = 0$$

$$V_o - 4I_D = 0$$

$$V_o = 4I_D = 4 \times (1.5 \text{ mA}) = 6 \text{ mV}$$

Diode Logic gates

R Gate

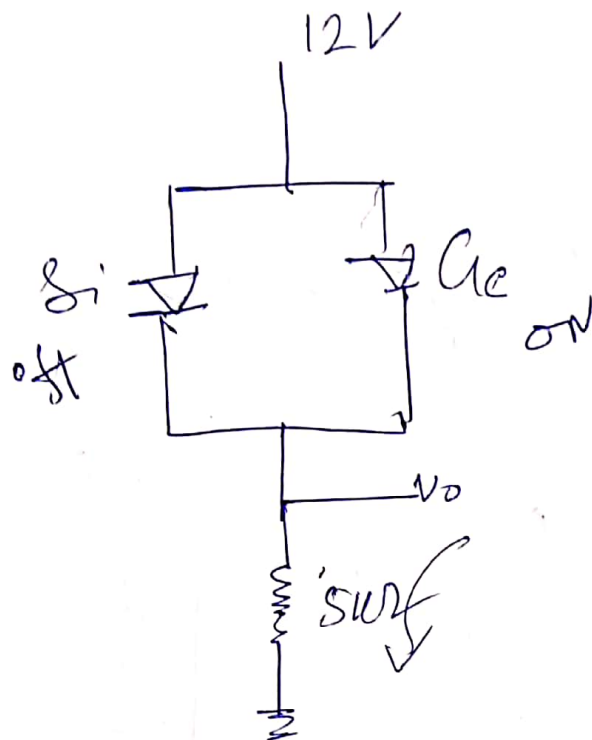


A	B	LED output
0	0	off
1	0	ON
0	1	ON
1	1	ON

0, = 0V, L
1 = 5V, H

6

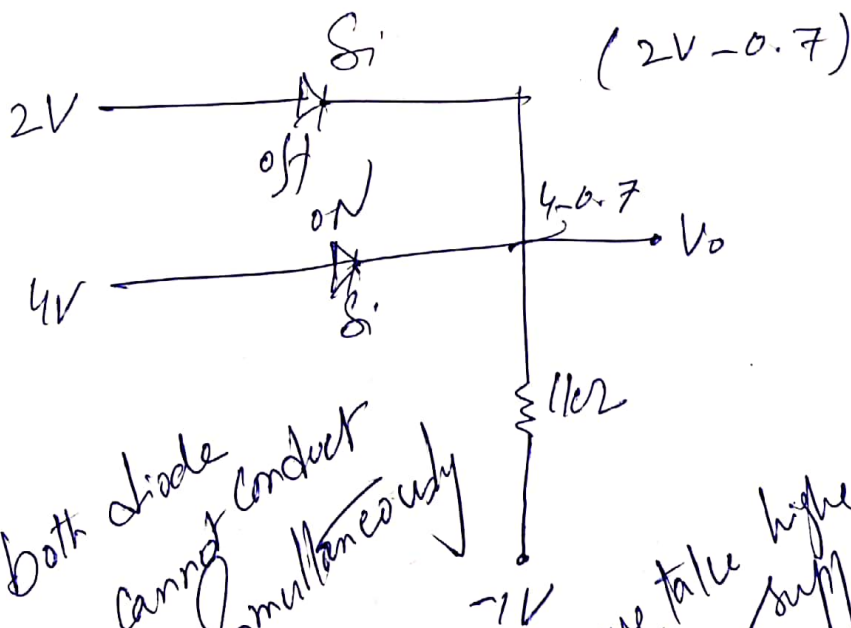
Q-10 Find V_o



$$V_o = 12 - 0.3 = 11.7V$$

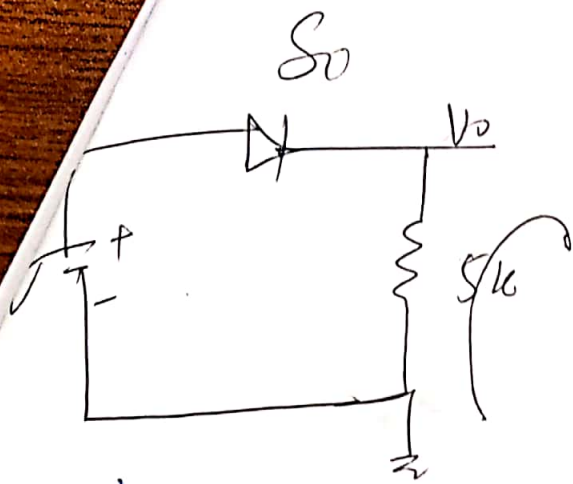
$$V_o = IR, \quad I = \frac{V_o}{R} = \frac{11.7}{5k\Omega}$$

Q-4



both diodes cannot conduct simultaneously

we take higher voltage then suppress and is conduct



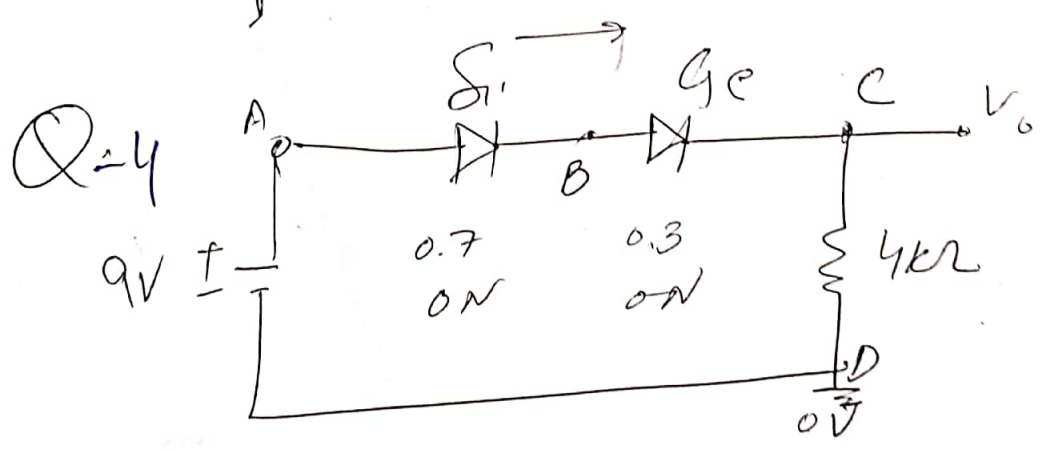
Q3

$$12 - 0.7 = 11.3 = V_0$$

Ans

$$I = \frac{V}{R} = \frac{11.3}{5k\Omega}$$

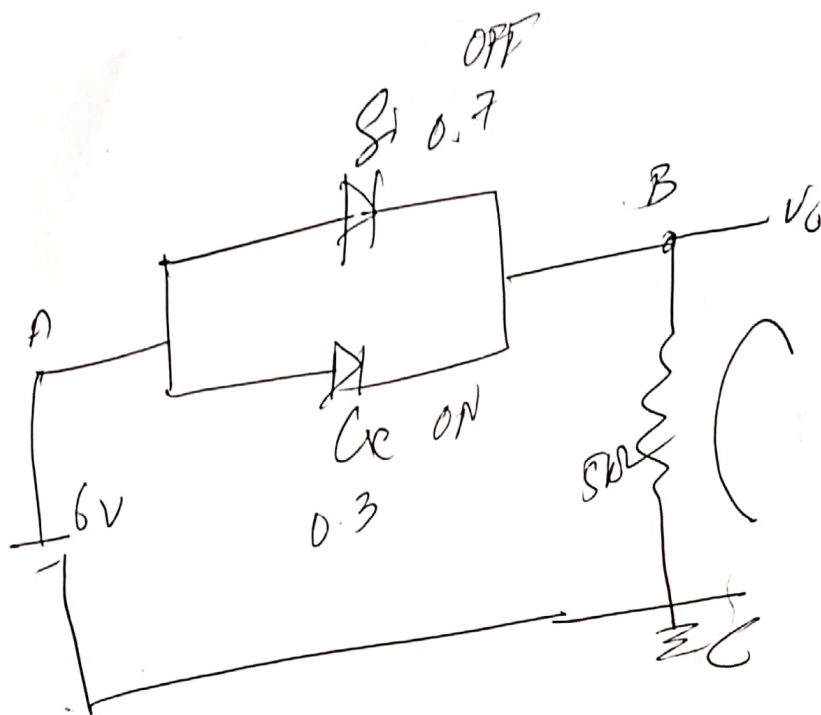
$$I = 2.26 \text{ mA}$$



$$9 - 0.7 - 0.3 = 4k\Omega I = 0$$

$$9 - 0.7 - 0.3 = V_0 \Rightarrow V_0 = 8V$$

$$I = \frac{8}{4k\Omega}$$

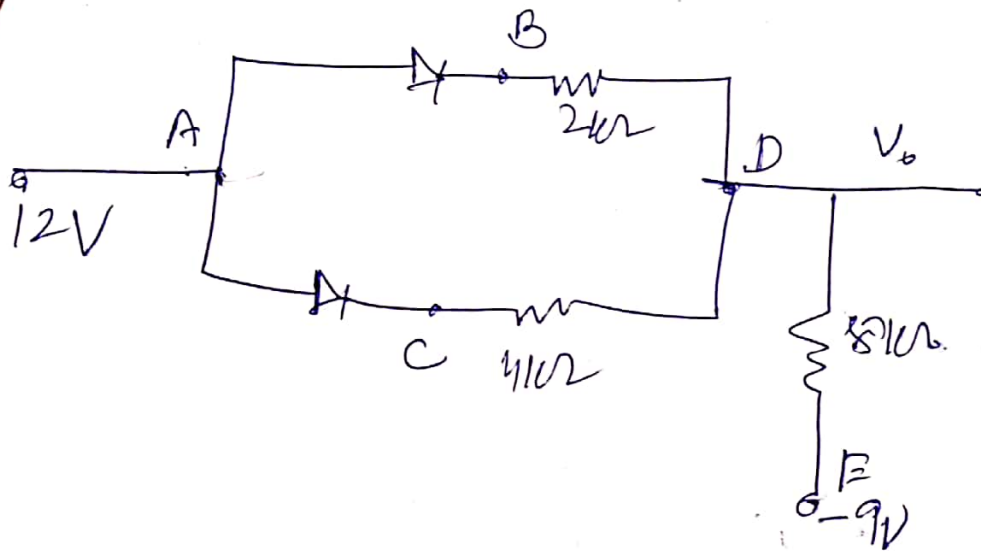


At B

$$6 - 0.3 = 5.7V$$

$$V_o = 5.7V$$

$$I = \frac{V}{R} =$$



$2k\Omega // 4k\Omega$ then R_{eq} is in series with $8k\Omega$

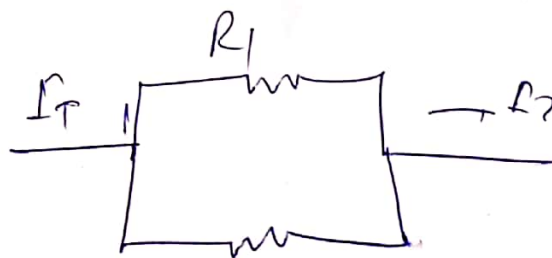
$$R_{eq} = \left(\frac{1}{2} + \frac{1}{4} \right)^{-1} = 1.33$$

$$R_T = 1.33k\Omega + 8k\Omega = 9.33k\Omega$$

$$I = \frac{\Delta V}{R_T} = \frac{11.3 + 9V}{9.33k\Omega}$$

$$I = 2.18mA$$

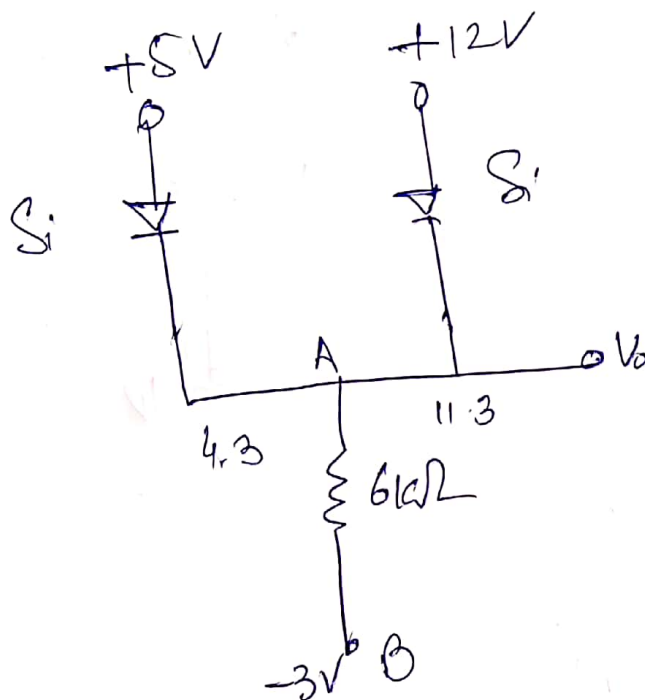
$$I_1 = \frac{4k\Omega (2.18mA)}{4k\Omega + 2k\Omega} = 1.48mA$$



$$I_2 = 0.73mA$$

(5)

Q-8



Solution:

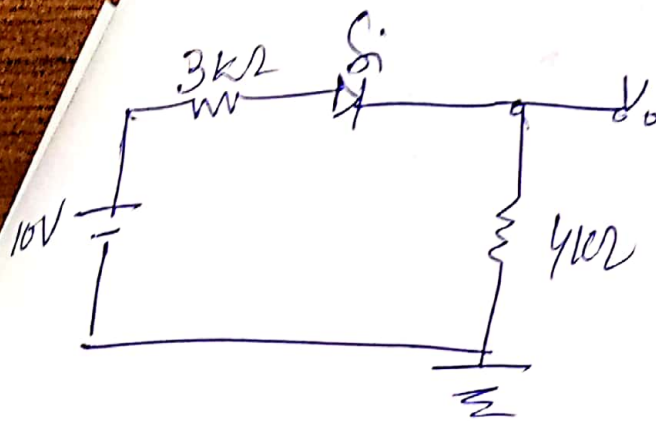
Since higher potential is to be than lower potential so $12V$ is considered - then lower potential more positive value we will get

So $V_o = \underline{\underline{11.3}}$

$$I = \frac{V}{R} = \frac{11.3 + 3V}{6k\Omega}$$

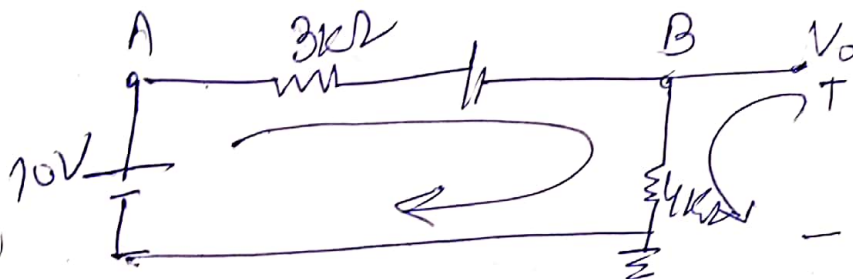
$$I = \frac{14.3}{6k\Omega} = 2.38 \text{ mA}$$

①



Solution :

$V_o = ?$



$$10 - (3k\Omega)I - 0.7 - (4k\Omega)I = 0$$

$$9.3 - 7I = 0$$

$$I = \frac{9.3}{7k\Omega} = 1.33 \text{ mA}$$

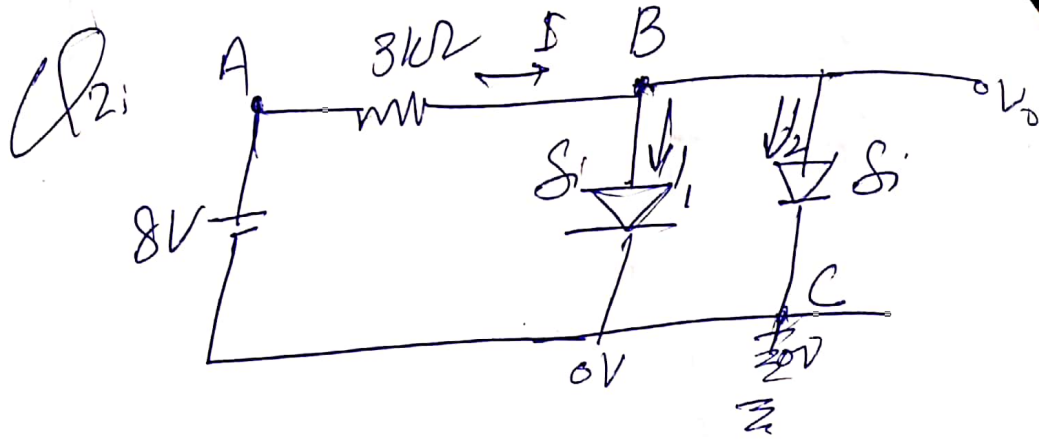
For 2nd loop

$$V_o - (4k\Omega)I = 0$$

$$V_o = (4k\Omega)(1.33 \text{ mA})$$

$$\boxed{V_o = 5.32 \text{ V}}$$

$$\frac{1.33}{4} = 0.332$$



Sol.

$$8V \xrightarrow{3k\Omega} 0.7$$

$$I = \frac{8 - 0.7}{3k\Omega} = \frac{7.3}{3k\Omega} = 2.43mA$$

if both diode equal/identical so

$$I = I_{D1} + I_{D2}$$

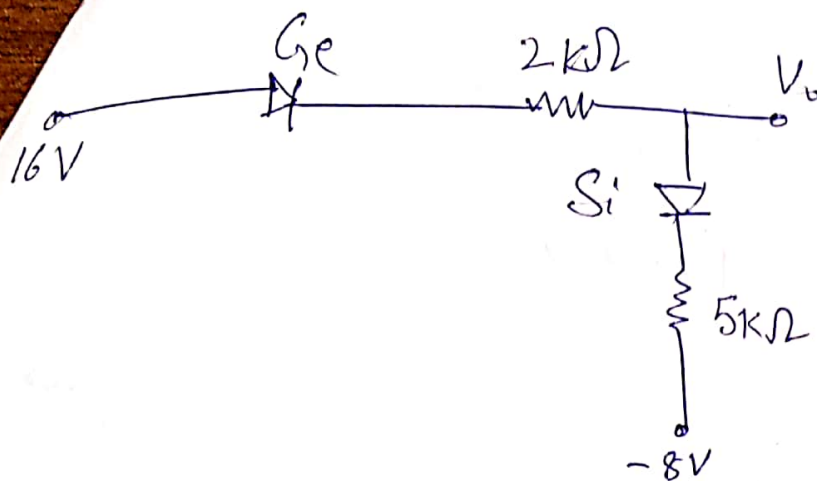
$$I = 2I_{D1}$$

$$I_{D1} = \frac{I}{2} = \frac{2.43}{2} = 1.215mA$$

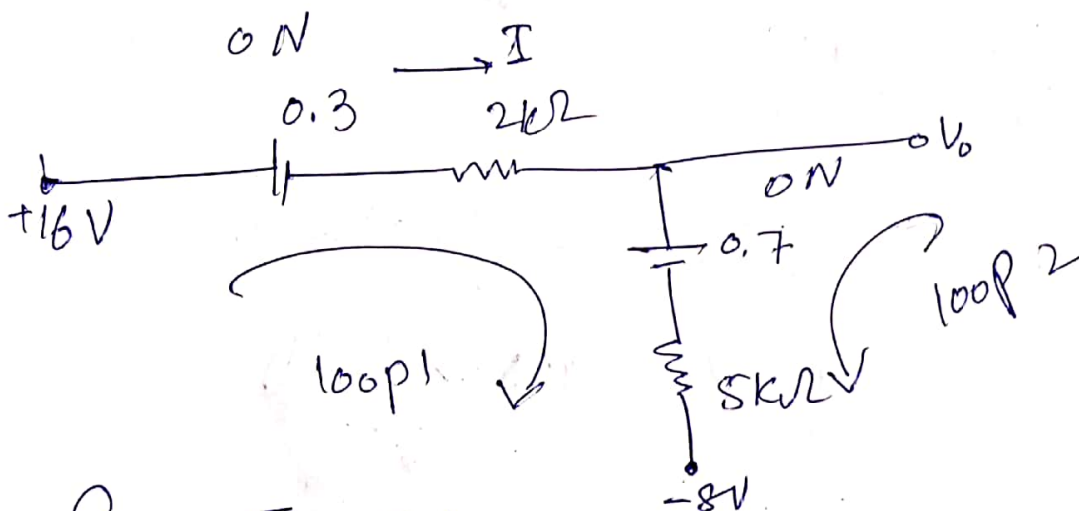
$$V_0 = 0.7V$$

Ans

(9)



Sol.

To find I , Take

loop 1

$$16 - 0.3 - I(2k\Omega + 5k\Omega) + 8 - 0.7 = 0$$

$$\frac{24 - 0.3 - 0.7}{7k\Omega} = I$$

7kΩ

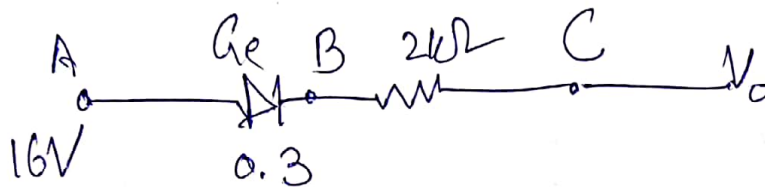
$$I = 3.286 \text{ mA}$$

For loop 2

$$-8 + (3.286 \text{ mA})(2 \text{ k}\Omega) + 0.7 = V_o$$

$$V_o = 9.13 \text{ V}$$

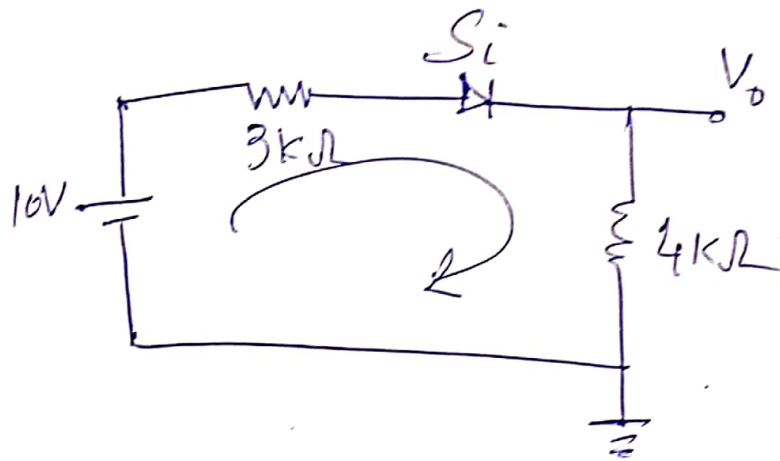
OR



$$16 - 0.3 - (3.286 \text{ mA})(2 \text{ k}\Omega) = V_o$$

$$V_o = 9.13 \text{ V} \quad \text{Ans}$$

Diode Problems (1)



Solution

$$10 - 0.7 = V_o$$

$$10 - V_{R_1} - 0.7 - V_{R_2} - V_o = 0$$

$$10 - 0.7 - I(3 + 4)k\Omega = V_o$$